

Course No. – AS 5570

Principles of Guidance for Autonomous Vehicles

Assignment 2

Due Date: September 25, 2025

(For Computer Assignments: Sept 30, 2025)

Part I : Deviated PP (DPP)

- 1) Consider a δ -angle Deviated Pure Pursuit (DPP)-guided pursuer with speed V_P against a maneuvering target with speed V_T and turn rate $\dot{\alpha}_T = c_T \dot{\theta}$. Consider three cases of $\nu = V_P/V_T \lesseqgtr 1$. For each of these three cases consider $c_T \lesseqgtr 1$. For each of these 9 cases:
 - i. Obtain the trajectory on the (V_θ, V_R) -space.
 - ii. In cases of successful capture:
 - i. What are the conditions of successful capture?
 - ii. Identify the capture region in terms of V_{R_0} and V_{θ_0} .
 - iii. Derive the expression of t_f .
 - iii. In other cases (no capture):
 - i. Derive the expressions of t_{miss} and R_{miss} .
 - iv. Derive expressions of $R(\psi)$ and $a_p(\psi)$. Obtain final values of R and a_p for $\mu \geq 1$, where $\nu \cos \delta = \mu \cos \psi_{CC}$, where ψ_{CC} is the value of ψ in collision course with speed ratio ν .
 - v. Obtain the value of $a_{p_{max}}$, and find at which value of ψ , $a_p(\psi) = a_{p_{max}}$.
 - vi. Compare the set of values of ν for PP and DPP for which the final value of a_p is finite.
- 2) Consider the initial engagement geometry in Question 3 on Page 121-122 of NPTEL lecture series. Consider speed ratios $\nu = 0.8, 1, 1.5$. For each of these three cases consider $c_T = 0, 0.8, 1, 1.5$, where $\dot{\alpha}_T = c_T \dot{\theta}$. Solve Question 3 with all these set-ups.

(Part A: A1, A2, A5 – **Computer assignment**; Part B: B1 – **Computer assignment**;
Part C: to be done after A1-A5 and B1 – **after Computer assignment**)

Part II : Line-Of-Sight (LOS) Guidance

- 3) NPTEL lecture series Questions 1 (A, B) and 2 A on Page 138-139 (Consider 10 kms of downrange and 1 km of altitude). (**Computer assignment**)

- 4) Consider an engagement geometry: Target's altitude $h = \text{constant}$, $\alpha_T = \pi$. Also, V_P, V_T are constants such that $V_P > V_T$. Suppose the pursuer is on the LOS guidance course and following LOS guidance command.

i. Derive the followings:

$$\text{i. } \left(\frac{dR_P}{d\theta} \right)^2 + R_P^2 = \frac{(\nu h)^2}{\sin^4 \theta}$$

$$\text{ii. } \cot \theta - \cot \theta_0 = -\frac{V_T}{h} t$$

$$\text{iii. } a_P = 2 \frac{V_P V_T}{h} \sin^2 \theta \left[1 + \frac{R_P \sin \theta \cos \theta}{\sqrt{(\nu h)^2 - (R_P \sin^2 \theta)^2}} \right]$$

ii. Show that

$$\text{i. } \alpha_{P_f} = \theta_f + \sin^{-1} \left(\frac{1}{\nu} \sin \theta_f \right)$$

$$\text{ii. } a_{P_f} = 2 \frac{V_P^2}{\nu h} \sin^2 \theta_f \left[1 + \frac{\cos \theta_f}{\sqrt{\nu^2 - \sin^2 \theta_f}} \right]$$

- iii. Let $V_P = 300$ m/sec, $V_T = 200$ m/sec, $h = 1500$ m, $\alpha_T = \pi$, $X_{P_0} = [0, 0]^T$ m, $X_{T_0} = [5000, 1500]^T$ m. It is also desired to satisfy the followings.

$$\text{i. } a_{P_f} < 6g, \text{ which approximately implies } 0 < \theta_f < 0.24\pi \text{ or } 0.6\pi < \theta_f < \pi$$

$$\text{ii. } t_f < 20 \text{ sec}$$

Then, what is the set of achievable impact angles (defined as $\alpha_{T_f} - \alpha_{P_f}$)?

- 5) Consider an LOS-guided pursuer with speed V_P be on LOS guidance course against a constantly maneuvering target such that its position is given as $X_T(t) = [c \cos \omega t, c \sin \omega t]^T$ and $V_P > c\omega$. Consider following three cases.

$$\text{i. } X_{P_0} = [0, 0]^T$$

$$\text{ii. } X_{P_0} = [-c, 0]^T$$

Obtain pursuer's trajectory: $R_P(\theta), X_P(t)$.

Part III : True Proportional Navigation (TPN) Guidance

- 6) Derive the conditions for successful capture of a non-maneuvering target by TPN ($a_P = -N'V_{R_0}\dot{\theta}$).
- 7) Consider an engagement between a non-maneuvering moving target and a TPN-guided pursuer ($a_P = c\dot{\theta}$).

- i. From engagement trajectories in (V_θ, V_r) -space, explain why the following conditions are necessary and sufficient for interception: (3)

$$\text{i. } V_{R_0} < 0,$$

- ii. $c > 0$,
 - iii. $V_{\theta_0}^2 + V_{R_0}^2 + 2cV_{R_0} < 0$.
- ii. Consider the following initial engagement geometry: $X_{P_0} = [0, 0]^T \text{m}$, $X_{T_0} = [1000, 0]^T \text{m}$, and $\alpha_{T_0} = 0$. Speeds of pursuer and target are: $V_p = 50 \text{m/sec}$ and $V_T = 15 \text{m/sec}$.
- i. Obtain the range of initial heading angles of the pursuer (α_{P_0}) that fall within the capture zone in terms of c .
 - ii. Obtain the minimum value of c to ensure a non-null capture zone.